

ASSEMBLYAI VOICE AGENT HACKATHON · SEPTEMBER 2026

constancia

An agent that phones your patients every week and remembers what they said last time.

constancia-voice.onrender.com · github.com/gmassello/constancia · MIT

About 7 in 10 patients in home rehabilitation abandon their exercise plan

70%

Not because the plan is wrong — because nobody follows up.

A physiotherapist with forty patients cannot make forty calls a week. So the follow-up does not happen, and by the time the patient comes back in, the month is lost.

The gap is not information. It is attention that scales.

THE PRODUCT IN ONE SENTENCE

constancia places the call, asks the protocol questions in a fixed order, escalates the moment a red flag shows up, writes a summary for the professional — and, the following week, **opens by asking about what the patient said last time.**

That last part is the whole product.

Week one, then week two

WEEK 1

- ¿Cuánto te duele hoy del uno al diez?
- La rodilla derecha me duele **siete de diez**, sobre todo cuando subo escaleras.

WEEK 2 · SAME QUESTIONS

- La semana pasada me contaste que la rodilla te dolía siete de diez al subir escaleras...
- Ahora me duele **cuatro de diez**.

~~pain 7/10 climbing stairs~~



pain 4/10 climbing stairs

the 7/10 is retired, not overwritten — it stays in the file with the dates it was true for

The professional opens the panel and sees a chain, not a number.

One call, six phases, no planner

recall

The file becomes the prompt, and her own words become the recogniser's key terms

converse

Four questions in declaration order, each answer through the red-flag guard

extract

Facts, each anchored to a literal quote from a turn that was hers

store

Contradictions retire the old fact. Nothing is ever deleted

Twilio · AssemblyAI Universal-Streaming v3 · Gemini Flash · ElevenLabs Flash v2.5 · Postgres + pgvector

The code decides, the model phrases.

Question order, the guard and the supersession are deterministic Python. The LLM writes sentences.

Why the memory claim is checkable, not asserted

1 No quote, no fact

Every stored fact carries a literal span of something the patient actually said, in a turn that was hers. Three conditions, eleven lines, and anything else is dropped.

2 Nothing is deleted

A contradiction retires the old fact; it never removes it. The professional can always see what was believed, when, and what replaced it.

3 The escalation is code

The red-flag guard is 32 deterministic lines with a negation window. The model phrases the escalation; it never decides on one.

Three verticals, one engine

Pack	Follows	Red flags
rehab	pain out of ten, sessions a week	sudden sharp pain, a fall, swelling with fever, numbness
postpartum	symptoms, mood	heavy bleeding
chronic	symptoms, a clinical reading, adherence	chest pain or breathlessness

The clinical content is configuration, not code. Adding a fourth vertical is writing a pack — nothing in the orchestrator, the extractor or the panel knows what rehab is.

The demo shows rehab. The engine is the same in all three.

The professional pays, not the patient

Who buys

A physiotherapist, a midwife, a chronic-care nurse — anyone with a caseload of people who should be doing something at home between appointments.

How it is sold

Monthly subscription per professional, tiered by active patients. Synthflow starts at US\$99 a month; a lower entry price aimed at Latin America is competitive.

Unit economics

Cents per call on AssemblyAI's streaming path — 333 free hours, then US\$0.15 an hour — against a 15-minute call the professional has no time to make.

Market

Not "physiotherapists in Argentina": any health professional doing longitudinal follow-up. One engine, many verticals.

ARGENTINA · LAW 25.326

Health data is sensitive data, and the demo treats it that way

Written informed consent, encryption in transit and at rest, registration as a data controller, breach notification within 72 hours.

The demo uses 100% fictitious patients. The product is designed for explicit consent before the first call — not a generic "I accept the terms".

What is actually built

319

tests green with no keys

The whole pipeline runs with no environment variable set and no database. The panel's buttons drive the real orchestrator against a scripted patient.

2

products, three capabilities

Universal-Streaming v3 live over the phone; Speech Understanding on the recording for entities, sentiment and key phrases. Feeding those phrases back to prime the next call was built, measured on a real recording, and reverted.

10

limits we say out loud

No authentication. One worker. Facts land seconds after the hangup, never during the call. Key phrases are shown, not fed back to the recogniser. The replay fixtures are synthetic and stay that way. The other five are in the README under *Honest limits* — every one of them written down rather than waited out.

Every number on this slide is checkable in the repository today.

Where each criterion is visible

Criterion	Where the judge sees it
Application of Technology	The live transcript on the panel, the key terms of each call, <code>app/stt.py</code> and <code>app/analysis.py</code>
Originality	The fact chain: the 7/10 struck through under the 4/10, each with the patient's own quote
Business Value	The public landing at <code>constancia-voice.onrender.com</code> , which states the case in plain language, and the pricing above
Presentation	Week 1, then week 2 with memory off, then week 2 with memory on — same patient

constancia — Spanish for keeping at it.